Confusion Matrix: Can you answer these 20 questions?

Machine Learning Interview Questions Part 12

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A confusion matrix is a table used to evaluate the performance of a classification model by comparing predicted labels with actual labels. It summarizes results into four key outcomes: true positives, true negatives, false positives, and false negatives. This structure shows not only how many predictions were correct, but also the types of errors the model made. From a confusion matrix, important metrics such as accuracy, precision, recall, and F1-score can be calculated. It is especially useful when dealing with imbalanced datasets, where accuracy alone can be misleading.

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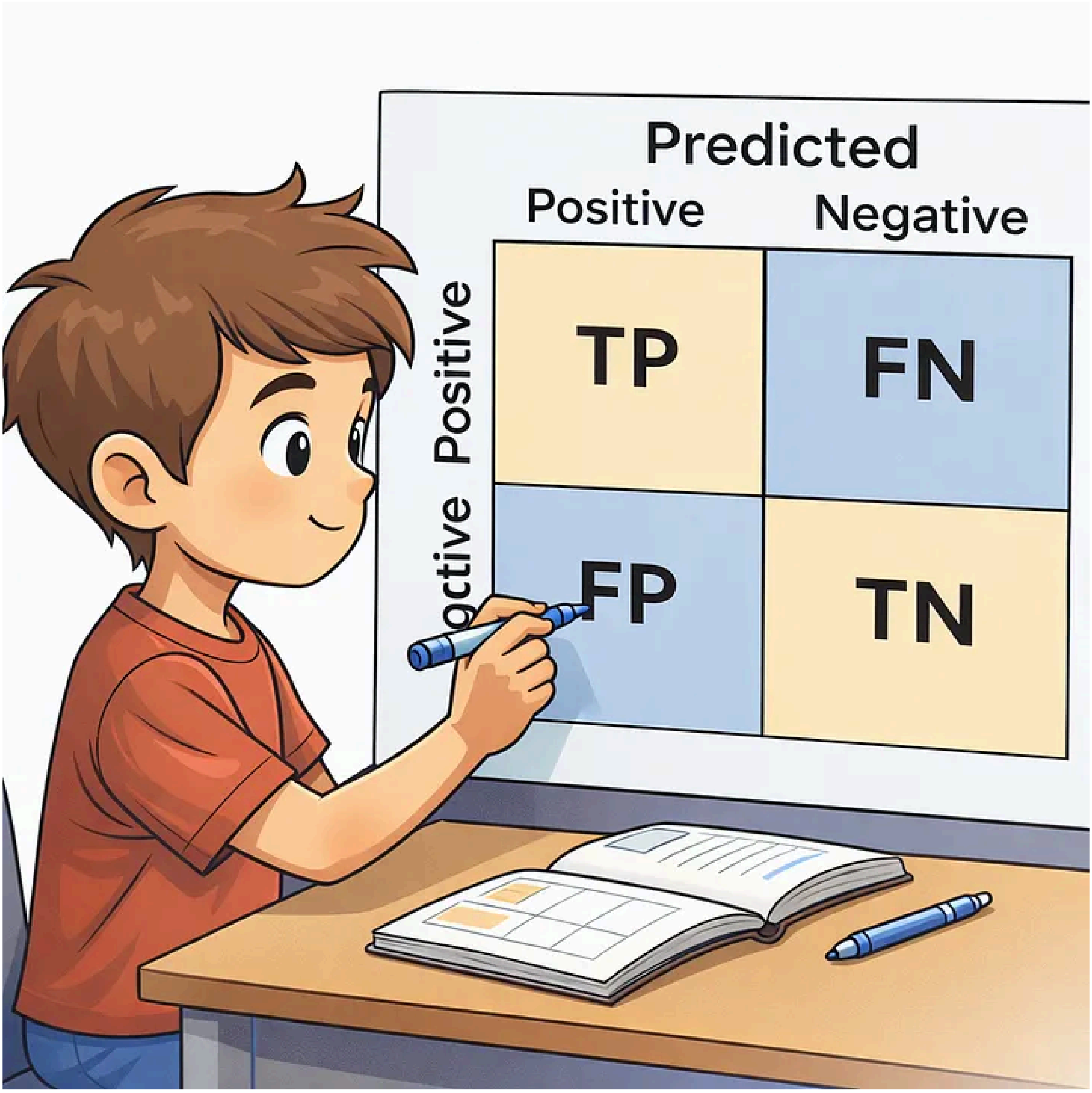

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Let's test your basic knowledge of confusion matrix. Here are 20 multiple-choice questions for you.

1. **Accuracy is simply a ratio of correctly predicted observations to the total observations. From the above confusion matrix, how would you define Accuracy?**

(A) Accuracy = (FP+FN)/(TP+FN+FP+TN)
(B) Accuracy = (TP+TN)/(TP+FN+FP+TN)
(C) Accuracy = (TP+FN)/(TP+FN+FP+TN)
(D) Accuracy = (FP+TN)/(TP+FN+FP+TN)

2. **From the confusion matrix, how would you define Error?**

(A) Error = (FP+FN)/(TP+FN+FP+TN)
(B) Error = (TP+TN)/(TP+FN+FP+TN)
(C) Error = (TP+FN)/(TP+FN+FP+TN)
(D) Error = (FP+TN)/(TP+FN+FP+TN)

3. **What's the correct definition for the True Positive Rate (TPR)? If P is the number of actual positives, N is the number of actual negatives, then total**

(B) TPR = TP/N = TP/(TP+FN)
(C) TPR = TP/P = TP/(TP+FN)
(D) TPR = TP/P = TP/(TP+FP)

4. **What's the correct definition for the fall-out or False Positive Rate (FPR)? Here, P is the number of actual positives, N is the number of actual negatives, and total population = P+N.**

(A) FPR = FP/P = FP/(FP+TN)
(B) FPR = FP/P = FP/(FP+TN)
(C) FPR = FP/N = FP/(FP+TP)
(D) FPR = FP/N = FP/(FP+TN)

5. **Recall is the ratio of correctly predicted positive observations to all observations in actual class. Recall = TP/P = TP/(TP+FN). Here, P is the number of actual positives. Recall is also known as -**

(A) Sensitivity, Hit Rate, True Positive Rate (TPR)
(B) Specificity, Hit Rate, True Positive Rate (TPR)
(C) Sensitivity, Miss Rate, True Positive Rate (TPR)
(D) Sensitivity, Hit Rate, Fall-out

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6. **What's the correct definition for Precision? (Select two)**

(A) Precision =TP/(TP+FP)
(B) Precision is the ratio of correctly predicted positive observations to the total predicted positive observations.
(C) Precision =FP/(TP+FP)
(D) Precision is the ratio of incorrectly predicted positive observations to the total predicted positive observations.

7. **Which statements about the Type-I and Type-II errors are correct? (Select two)**

(A) Type-I error = FPR
(B) Type-II error = FNR
(C) Type-I error = FNR
(D) Type-II error = FPR

8. **The F1 score is the harmonic mean of Precision and Recall. What's the correct formula for the F1 score?**

(A) F1 score = (2*Precision*Recall)/(Precision + Recall)
(B) F1 score = (2*Precision*TPR)/(Precision + TPR)

(C) F1 score = (2*Precision*Sensitivity)/(Precision + Sensitivity)
(D) All of the options

9. Which statement about the Receiver Operating Characteristic (ROC)-(Area Under the Curve) AUC Curve is correct?

(A) ROC is a probability curve that plots the true positive rate (sensitivity or recall) against the false positive rate (1 — specificity) at various thresholds.
(B) AUC is the area under the ROC curve. If the AUC is high (close to 1), the model is better at distinguishing between positive and negative classes.
(C) If AUC = 0.5, it represents a model that is no better than random.
(D) All of the options.

10. Which statement about the ROC-AUC Curve is correct?

(A) Thresholding is used to create binary classification outcomes. For example, threshold = 0.5, if probability ≥ 0.5, then predicted class = 1, and if probability < 0.5, then predicted class = 0. By changing this threshold, you can increase or decrease the recall or precision of a regressor, and this trade-off is visualized with a ROC curve.
(B) Thresholding is used to create binary classification outcomes. For example, threshold = 0.5, if probability < 0.5, then predicted class = 1, and if probability > 0.5, then predicted class = 0. By changing this threshold, you can increase or decrease the recall or precision of a classifier, and this trade-off is visualized with a ROC curve.
(C) Thresholding is used to create binary classification outcomes. For example, threshold = 0.5, if probability ≥ 0.5, then predicted class = 1, and if probability < 0.5, then predicted class = 0. By changing this threshold, you can increase or decrease the recall or precision of a classifier, and this trade-off is visualized with a ROC curve.
(D) Thresholding is used to create binary classification outcomes. For example, threshold = 0.5, if probability ≥ 0.5, then predicted class = 1, and if probability < 0.5, then predicted class = 0. Without changing this threshold, you can increase or decrease the recall or precision of a classifier, and this trade-off is visualized with a ROC curve.

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11. You are analyzing a binary classification model built for an online retail platform to predict whether a user will complete a purchase based on their browsing activity. Only a small fraction of users actually make a purchase, resulting in a highly imbalanced dataset. Because of this imbalance, relying on a single metric may provide a misleading view of model performance. Your goal is to evaluate how effectively the model identifies purchasing customers while also understanding the trade-offs between false positives and false negatives. Which two evaluation methods or metrics should be prioritized to obtain a meaningful and well-rounded assessment of the model's performance? (Select two.)

(A) Rely on overall accuracy as the main performance metric.
(B) Use Root Mean Squared Error (RMSE) to quantify the distance between the predicted probability and the binary outcome.
(C) Use precision and recall to emphasize correct identification of purchasing customers while accounting for false positives and false negatives.
(D) Analyze the confusion matrix to understand true positives, false positives, true negatives, and false negatives, and to derive metrics such as precision, recall, and F1 score.

12. A risk analytics team is building a binary classifier to estimate the likelihood that borrowers will miss repayments. Historical data show that only a small fraction of customers default, making the target variable highly skewed. To properly assess how well the model separates high-risk and low-risk customers, the team evaluates it using the ROC curve and the Area Under the Curve (AUC) rather than accuracy alone. Which statement most accurately describes how ROC and AUC should be interpreted in this setting?

(A) A ROC curve near the diagonal line with an AUC around 0.5 indicates strong predictive performance across thresholds.
(B) An AUC value close to 1.0 means the model has a strong ability to distinguish between defaulting and non-defaulting customers.
(C) An AUC close to 0.7 shows the classifier is highly effective at predicting both

classes.
(D) A ROC curve near the top-left corner implies the model is overfitting and producing unrealistically optimistic predictions.

13. **A medical analytics team maintains a binary classifier that flags patients who may develop a certain illness. A revised version of the model is being prepared, and the engineer must recalibrate it so that correct predictions are maximized for both high-risk patients and low-risk patients. The evaluation metric should directly reflect overall correctness across both classes. Which metric is most appropriate for this recalibration task?**

(A) Accuracy
(B) Precision
(C) Recall
(D) Root mean squared error (RMSE)

14. **You are developing a supervised learning model for a medical screening system that outputs a binary decision indicating whether a patient is likely to have a serious condition. Both types of errors are costly: incorrectly flagging healthy patients leads to unnecessary diagnostic procedures, while missing true cases may delay life-saving treatment. To responsibly assess the model before release, which evaluation metrics should be emphasized to appropriately manage both risks? (Select two)**

(A) Emphasize overall accuracy because it summarizes the proportion of correct predictions across the dataset.
(B) Focus on precision to limit the number of healthy patients incorrectly classified as positive.
(C) Focus on recall to ensure most patients who truly have the condition are detected.
(D) Use the F1 score to capture the balance between precision and recall.

15. **You are developing a predictive system for a financial institution to decide whether customer applications for credit should be accepted or declined. The model uses inputs such as monthly earnings, repayment history score, and**

years of work experience. The institution is concerned not only with predictive quality but also with ensuring that outcomes are equitable across sensitive attributes like age group, sex, or background. You are asked to choose evaluation techniques that capture both predictive effectiveness and potential unfair treatment across groups. Which option best satisfies these goals?

(A) Assess the model using accuracy and AUC-ROC, and inspect whether input variable distributions differ across demographic categories.
(B) Evaluate the model using F1 score and AUC-ROC, and verify whether true positive rates are comparable across different demographic groups.
(C) Use accuracy as the sole performance metric and review feature importance to confirm reliance on meaningful variables.
(D) Measure precision and recall independently and test for bias by checking correlations between sensitive attributes and predictions.

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16. **A medical research team is creating a predictive model to identify patients who may develop a rare but life-threatening condition. Failing to identify an affected patient could delay intervention and cause serious harm. In contrast, incorrectly flagging a healthy patient may lead to additional diagnostic tests, which are comparatively less harmful and manageable. Which model performance characteristic should the team emphasize to align with these priorities?**

(A) Prioritize a model with high precision
(B) Prioritize a model with high accuracy
(C) Prioritize a model with low recall
(D) Prioritize a model with high recall

17. **A medical analytics firm is building a predictive model to identify patients who may develop a rare but severe condition. Only a very small fraction of records represent positive cases. The solution must correctly detect as many high-risk patients as possible while also limiting incorrect positive predictions that**

could lead to unnecessary tests and anxiety. Which evaluation metric is most appropriate for assessing this model's performance?

(A) Accuracy
(B) Precision
(C) Recall
(D) F1 Score

18. **A machine learning analyst built two binary classifiers to predict whether a person carries a certain virus. Positive means infected; negative means virus-free. A contract defines penalties for model errors:**

- **If the model predicts infection for a virus-free person, there is a $200 penalty (false positive).**
- **If the model predicts virus-free for an infected person, there is a $3,000 penalty (false negative).**

Confusion matrices on the same test set are:

Model A: TP=560, FN=60, FP=100, TN=1200
Model B: TP=520, FN=100, FP=90, TN=1210

Which model should be selected to minimize total penalty, and why?

(A) Model A because it has the highest recall.
(B) Model B because it has the highest recall.
(C) Model A because its total penalty cost is lower than Model B's.
(D) Model B because its total penalty cost is lower than Model A's.

19. **A data scientist developed an image classification model that assigns images to one of ten classes. The model produced the following performance metrics:**

Accuracy on training data: 95%
Accuracy on validation data: 79%
Accuracy on test data: 77%

Assume a subject-matter expert can achieve 97% accuracy on the same task. Which conclusion best describes the model's behavior?

(A) The model has high bias and low variance.
(B) The model has low bias and high variance.
(C) The model has high bias and high variance.
(D) The model has low bias and low variance.

20. **A machine learning practitioner trained a neural network and observed the following results:**

Training accuracy: 96%
Validation accuracy: 55%
Test accuracy: 51%

There is a significant performance gap between the training data and both evaluation datasets. Which of the following are plausible reasons for this behavior? (Select two)

(A) The model is overfitting the training data.
(B) The training process ended too early and requires many more epochs.
(C) The model architecture is too shallow and must include more layers.
(D) The validation and test datasets follow a different data distribution from the training set.

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The solutions will be found here:

Top 20 Confusion Matrix Interview Questions and Answers (Part 1)
Top 20 Confusion Matrix Interview Questions and Answers (Part 2)

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